

24×432-Gbps/λ All-Optical Equalization for DSP-Free Optical Interconnects

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Abstract We demonstrate an integrated all-optical signal processor (OSP), enabling WDM transmission of 24-channel at 432-Gbps/λ over 2-km SMF across the whole C-band. The OSP simultaneously equalizes fiber-dispersion and transceiver-bandwidth-limit with <50-mW power and <75-ps latency, offering a disruptive solution for energy-efficient and low-latency optical interconnects. © 2026 The Author(s)

Introduction

The explosive growth of large-scale artificial intelligence (AI) clusters has placed unprecedented demands on optical interconnects, particularly concerning shoreline density and power efficiency [1]. In modern distributed machine learning architectures, microsecond-level latency can severely bottleneck graphics processing unit (GPU) synchronization and parallel computing efficiency [2]. As optical interconnects scale to exceed 400 Gbps per lane, conventional digital signal processing (DSP) algorithms must become increasingly complex, leading to substantial power consumption and significant processing latency [3]. Moreover, in intensity-modulation/direct-detection (IM/DD) systems, DSP-based equalization inherently loses phase information, limiting its ability to efficiently compensate for chromatic-dispersion (CD)-induced power fading [4]. While nonlinear compensation can be performed in the digital domain, it requires excessive computational complexity, further hindering the mitigation of transceiver bandwidth limits (BWL) and nonlinearities [5]. These limitations not only restrict transmission distance and reduce the usable wavelength-division multiplexing (WDM) window but also constrain the achievable aggregate data rate across multiple wavelengths [6-13]. The narrow usable WDM window also restricts the deployment of uncooled DFB lasers, particularly for transmission distances over 2 km [14].

Consequently, the transition towards all-optical equalization [15-19] or hybrid optical/electrical equalization [20, 21] transceiver architectures has become a critical objective.

In this work, we experimentally demonstrate an integrated all-optical signal processor (OSP) based on a multi-tap infinite impulse response (IIR) filter and a lattice-form finite impulse response (FIR) filter. Our OSP enables a record single-lane data rate of 432 Gbps and an aggregate throughput of 10.36 Tbps over 2 km of single-mode fiber (SMF) using 24 wavelengths across the entire C-band, by exceeding the performance of digital equalizers (Fig. 1). Crucially, we show that the OSP can simultaneously compensate for fiber CD and system bandwidth limitations, yielding a bit error rate (BER) even below the back-to-back (BtB) baseline without bandwidth compensation. With power consumption below 50 mW and processing latency under 75 ps, this OSP provides a highly viable path toward future high-throughput and energy-efficient data center interconnects.

OSP architecture and implementation

The OSP is an integrated photonic circuit designed as a programmable, universal equalizer capable of mitigating both linear and nonlinear distortions. Building upon our previous reservoir-based designs [15], we refine the architecture to optimize optical loss and ensure compatibility with high-baud-rate, high-order modulation signals.

As illustrated in Fig. 2(a), the photonic circuit comprises a multi-tap IIR filter followed by a lattice-form FIR filter. The photonic IIR stage functions as a temporal memory unit, where multi-path recursive feedback creates an effective memory window that extends beyond physical delays while minimizing recursive losses. This stage utilizes three physical delays τ_1 , τ_2 , and τ_3 of 15 ps, 5 ps, and 15 ps, respectively. By tuning the integrated phase shifters, these delays can construct four distinct path delays, significantly

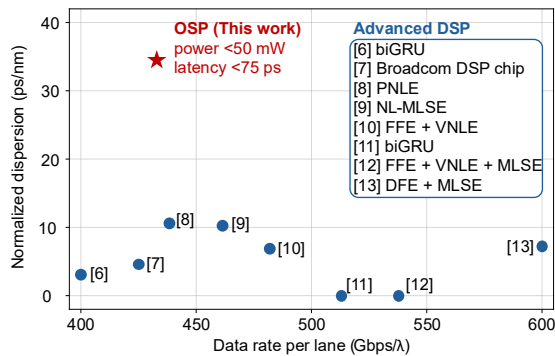


Fig. 1: Comparison of over 400-Gbps IM/DD transmission using our OSP and advanced DSP [6-13].

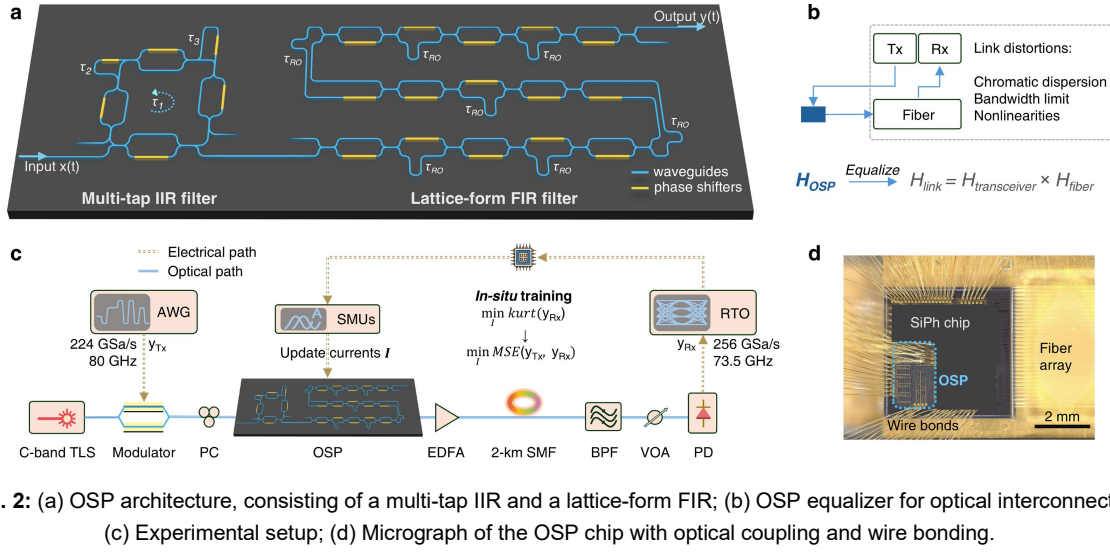


Fig. 2: (a) OSP architecture, consisting of a multi-tap IIR and a lattice-form FIR; (b) OSP equalizer for optical interconnects; (c) Experimental setup; (d) Micrograph of the OSP chip with optical coupling and wire bonding.

broadening the processing window. Complementing the photonic IIR stage, the photonic readout layer has been redesigned as a lattice-form FIR filter consisting of cascaded unbalanced Mach-Zehnder interferometers (UMZIs). Each UMZI maintains a differential delay of $\tau_{RO} = 4$ ps between the two arms. Seven UMZIs introduce a delay increasing incrementally from 0 to $7\tau_{RO}$. By optimizing the phase shifters, the overall insertion loss is only 3 dB, thereby preserving the signal-to-noise ratio for high-order modulation formats. Crucially, the OSP operates on the complex optical field prior to square-law photodetection, so it can perform phase-sensitive compensation that is physically impossible for post-detection DSP in IM/DD systems. The OSP is fabricated on a standard silicon-on-insulator (SOI) platform, ensuring scalability via high-volume complementary metal-oxide-semiconductor (CMOS) manufacturing processes.

Fig. 2(c) shows the experimental setup. At the transmitter, communication signals are generated using an 80-GHz arbitrary waveform generator (AWG) and directly modulated using a 100-GHz thin-film lithium niobate (TFLN) intensity modulator. A raised-cosine filter with a roll-off factor of 0.01 is employed to improve spectral efficiency. Clipping is applied to optimize the peak-to-average power ratio, and no electrical pre-emphasis is used during signal generation. Our OSP chip, shown in Fig. 2(d), is deployed at the transmitter side and serves as an all-optical pre-compensator. An erbium-doped fiber amplifier (EDFA) then boosts the optical power to 12 dBm, after which the pre-distorted signal propagates through 2 km of SMF. At the receiver, a band-pass filter (BPF) suppresses out-of-band noise. A variable optical attenuator (VOA) is used to maintain the received power at 6 dBm before the signal is detected by a 70-GHz photodetector (PD)

and captured by a 73.5-GHz real-time oscilloscope (RTO). The on-chip parameters (i.e., currents) are optimized for different signals using source/measure units (SMUs) with a two-stage approach. We first minimize the kurtosis of the output signal y_{Rx} to accelerate convergence and then minimize the mean-square error between y_{Rx} and the target signal y_{Tx} to improve signal quality.

Experimental results and discussion

To evaluate the OSP's efficacy against chromatic dispersion and transceiver bandwidth limitations, we perform high-speed transmission experiments over a 2-km SMF link across the C-band. We test signals of PAM4 and PAM8 formats at eight symbol rates ranging from 100 to 160 Gbaud, thereby demonstrating the chip's versatility under a wide range of operating conditions. The maximum achievable baud rate is primarily constrained by the bandwidth of the characterization equipment. For each configuration, on-chip phase shifters are optimized using a genetic algorithm (GA) to minimize the BER.

Fig. 3(a) compares the BER performance of BtB transmission with 2-km SMF transmission utilizing the OSP at 1550 nm. The OSP successfully supports up to 136 Gbaud PAM4 and 120 Gbaud PAM8 signals below the 7% overhead (OH) hard-decision forward error correction (HD-FEC) threshold without requiring auxiliary DSP equalizers. Furthermore, 152 Gbaud PAM4 and 136 Gbaud PAM8 signals satisfy the 20% soft-decision FEC (SD-FEC) requirement, while 160 Gbaud PAM4 and 144 Gbaud PAM8 signals remain within the 25% SD-FEC limit. For PAM8 signals exceeding 144 Gbaud, the BERs inevitably degrade due to the Nyquist limit. These results confirm that the proposed all-optical equalization approach can sustain ultra-high per-lane rates

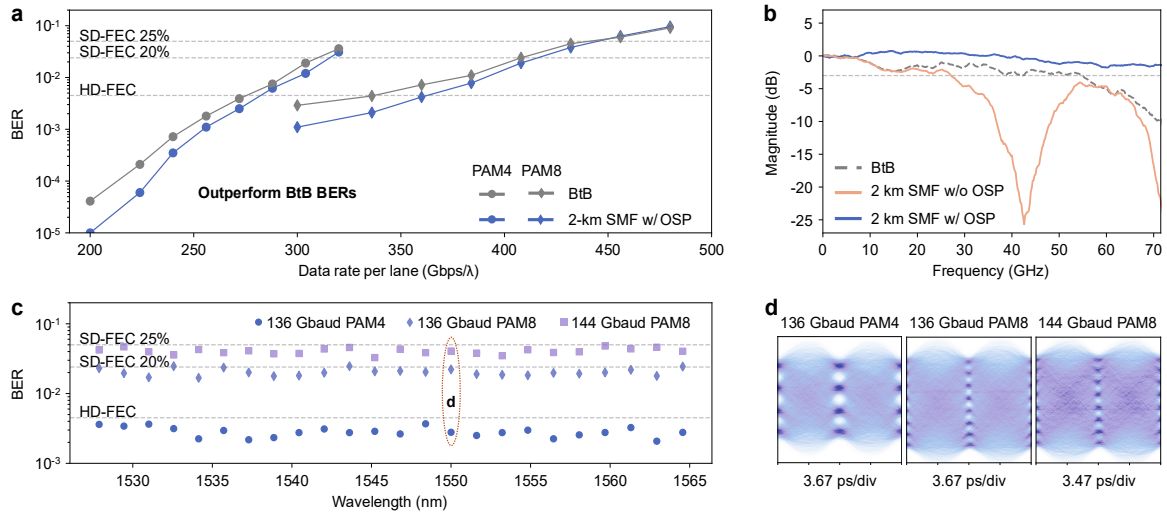


Fig. 3: (a) Measured BER as a function of data rates for PAM4 and PAM8 signal transmission. (b) The spectral responses after detection of different situations. (c) Measured BER as a function of wavelength for 136 Gbaud PAM4 and PAM8, and 144 Gbaud PAM8. (d) Equalized eye diagrams of 2-km SMF transmission with OSP for different signals at 1550 nm.

under practical short-reach transmission conditions.

Notably, the BER after 2-km transmission with OSP equalization outperforms the BtB results obtained without the OSP. This "Beyond BtB" performance is attributed to the OSP's ability to simultaneously equalize transceiver bandwidth limitations and the specific power fading notches induced by fiber dispersion over 2 km. Fig. 3(b) depicts the frequency domain channel responses for BtB (grey), 2-km SMF without OSP (yellow), and 2-km SMF with OSP (blue). By reshaping the response directly in the optical domain, the OSP not only restores a flatter channel response than that of the original BtB link, but also effectively extends the link bandwidth beyond 70 GHz. Consequently, the bandwidth requirements for the transceiver's electrical and optoelectronic components are substantially relaxed. At very high symbol rates, however, the BER gap between the two cases narrows because the system gradually approaches a noise-limited regime.

We further investigate the broadband operational compatibility of the OSP by individually optimizing the chip for different wavelengths covering the entire C-band (1525 nm to 1565 nm) with 200-GHz channel spacing. The performance is verified using 136 Gbaud PAM4, 136 Gbaud PAM8, and 144 Gbaud PAM8 signals over 2 km of SMF. As shown in Fig. 3(c), consistent BER performance across the entire 40-nm wavelength range is maintained for each transmitted signal. Furthermore, all 24 channels support 136 Gbaud PAM4 and PAM8 transmission below the 7% OH HD-FEC threshold and 20% SD-FEC threshold, respectively. Specifically, we achieve a record aggregate throughput of 10.36 Tbps, with all 24 wavelengths supporting 144 Gbaud PAM8 (432 Gbps per channel) under the 25% SD-FEC

threshold. The corresponding eye diagrams in Fig. 3(d) further confirm that link distortions induced by severe CD and bandwidth limit are effectively equalized by the OSP.

Ultimately, despite this high performance, the OSP maintains a total power consumption below 50 mW, which is orders of magnitude lower than comparable CMOS-based DSPs. By implementing air-trenched thermal phase shifters, the power consumption can be further reduced to below 5 mW. Furthermore, the processing latency remains under 75 ps, satisfying the stringent requirements for low-latency AI fabrics.

Conclusions

We experimentally demonstrate an integrated all-optical signal processor that supports record single-lane transmission rates of up to 432 Gbps and an aggregate throughput of 10.36 Tbps over 2 km of SMF using 24 wavelengths across the entire C-band. Notably, the OSP achieves "beyond-BtB" performance by simultaneously mitigating transceiver bandwidth limitations and fiber-dispersion-induced power fading, directly on the optical field. With sub-50-mW power consumption and sub-75-ps latency, the proposed OSP represents a compelling low-power and low-latency solution for future high-density optical interconnects.

Acknowledgements

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References

- [1] SemiAnalysis, "Co-Packaged Optics (CPO) Book – Scaling with Light for the Next Wave of Interconnect", 2026, <https://newsletter.semianalysis.com/p/co-packaged-optics-cpo-book-scaling>.
- [2] SemiAnalysis, "Multi-Datacenter Training: OpenAI's Ambitious Plan To Beat Google's Infrastructure", 2024, <https://semianalysis.com/2024/09/04/multi-datacenter-training-openais/>.
- [3] OIF, "Next Generation CEI-448G Framework", 2025, oiforum.com/wp-content/uploads/OIF-FD-CEI-448G-01.0.pdf.
- [4] M. Chagnon, "Optical communications for short reach", *Journal of Lightwave Technology*, vol. 37, no. 8, pp. 1779–1797, 2019. DOI: [10.1109/JLT.2019.2901201](https://doi.org/10.1109/JLT.2019.2901201).
- [5] D. Che, X. Chen, "Modulation format and digital signal processing for IM-DD optics at post-200G era," *Journal of Lightwave Technology*, vol. 42, no. 588, pp. 588–605, 2023. DOI: [10.1109/JLT.2023.3311716](https://doi.org/10.1109/JLT.2023.3311716).
- [6] C. Han, Q. Yang, J. Qin, Y. Zhou, Z. Zheng, Y. Zhang, H. Wang, Y. Sun, J. Lu, Y. Wang, Z. Ge, Y. Wu, L. Wang, Z. He, S. Yu, W. Hu, C. Peng, H. Shu, J. E. Bowers, and X. Wang, "Exploring 400 Gbps/λ and beyond with AI-accelerated silicon photonic slow-light technology," *Nature Communications*, vol. 16, no. 1, p. 6547, 2025. DOI: <https://doi.org/10.1038/s41467-025-61933-5>.
- [7] BROADCOM, "BCM83640-DIE 3-Nm CMOS 1.6T (8:4) PAM-4 Transceiver PHY", 2026, <https://docs.broadcom.com/doc/83640-DIE-PB>.
- [8] C. St-Arnault, S. Bernal, E. Berikaa, Z. Wei, R. Gutiérrez-Castrejón, J. Zhang, M. S. Alam, A. Nikic, B. Krueger, F. Pittalà, and D. V. Plant, "Net 1.6 Tbps (4×400Gbps/λ) O-Band IM/DD Transmission Over 2 km Using Uncooled DFB Lasers on the LAN-WDM grid and Sub-1V Drive TFLN Modulators," in *Optical Fiber Communication Conference (OFC) 2024* (Optica Publishing Group, 2024), p. Th4C.6.
- [9] H. Taniguchi, S. Yamamoto, M. Nakamura, Y. Nakajima, and E. Yama-zaki, "Net 400-Gb/s/lane O-band IM-DD Transmission Using 182-GBd PAM-6 with KP4+SFEC over 20-km SSMF," in *2025 European Conference on Optical Communications (ECOC)* (IEEE, 2025), pp. 1–4.
- [10] L. Kulmer, T. Blatter, M. Kohli, Y. Horst, S. M. Koepfli, and J. Leuthold, "Single Carrier net 400 Gbit/s IM/DD over 400 m Fiber enabled by Plasmonic Mach-Zehnder Modulator," in *Optical Fiber Communication Conference (OFC) 2024* (Optica Publishing Group, 2024), p. W4H.5.
- [11] Y. Zhang, H. Shu, Y. Guo, P. Zhou, L. Wang, J. Cai, L. Yao, L. Yang, L. Li, T. Long, Z. Zhang, C. Han, K. Lu, Y. Sun, Z. Xu, J. Qin, Y. Tong, Z. He, X. Xiao, L. Wang, B. Chen, S. Yu, and X. Wang, "Integrated photonics enabling ultra-wideband fiber–wireless communication," *Nature*, vol. 651, pp. 348–355, 2026. DOI: <https://doi.org/10.1038/s41586-026-10172-9>.
- [12] X. Chen, P. Liao, F. Tang, Z. Liu, H. Wang, R. Cronin, Y. Zhao, H. Zhang, J. Moss, S. Rihani, J. Griffiths, D. Drysdale, L. Regemorter, X. Liu, D. Childs, and P. Aivaliotis, "540Gbps IMDD Transmission over 30km SMF using 110GHz Bandwidth InP EML," in *Optical Fiber Communication Conference (OFC) 2025* (Optica Publishing Group, 2025), p. Th4B.2.
- [13] C. St-Arnault, S. Bernal, D. Kita, R. Dickson, M. Y. Abdelaziz, A. Nikic, B. Qiu, B. Krueger, F. Pittalà, C. Reimer, N. Ben-Hamida, and D. V. Plant, "Net 3.2 Tbps 225 Gbaud PAM4 O-Band IM/DD 2 km Transmission Using FR8 and DR8 with a CMOS 3 nm SerDes and TFLN Modulators," in *Optical Fiber Communication Conference (OFC) 2025* (Optica Publishing Group, 2025), p. Th4B.1.
- [14] C. St-Arnault, S. Bernal, E. Berikaa, Z. Wei, R. Gutiérrez-Castrejón, J. Zhang, M. S. Alam, A. Nikic, B. Krueger, F. Pittalà, and D. V. Plant, "Practical Fiber Dispersion-Induced Limitations for 1.6 Tbps (4×400Gbps/λ) O-Band IM/DD Transmission Systems Over 2, 10, 20 and 40 km," *Journal of Lightwave Technology*, vol. 43, no. 7, pp. 3222–3232, 2025, DOI: [10.1109/JLT.2024.3519901](https://doi.org/10.1109/JLT.2024.3519901).
- [15] B. Wang, Q. Xiao, T. Xu, L. Fan, S. Liu, J. Dong, J. Zhang, and C. Huang, "Beyond Terabit/s Integrated Neuromorphic Photonic Processor for DSP-Free Optical Interconnects," *arXiv:2504.15044*, 2025. <https://arxiv.org/pdf/2504.15044>.
- [16] R. Van Assche, S. Masaad, E. Gooskens, S. Sackesyn, J. Van Kerrebrouck, X. Yin, and P. Bienstman, "Real-time all-optical signal equalisation with silicon photonic recurrent neural networks," in *2025 European Conference on Optical Communications (ECOC)* (IEEE, 2025), pp. 1–4.
- [17] E. Staffoli, M. Mancinelli, P. Bettotti, and L. Pavesi, "Equalization of a 10 Gbps IMDD signal by a small silicon photonics time delayed neural network," *Photonics Research*, vol. 11, no. 5, pp. 878–886, 2023. DOI: <https://doi.org/10.1364/PRJ.483356>.
- [18] L. Zhang, T. Wu, L. Li, and M. Xiao, "Dispersion Managed Transceiver Extending CWDM IMDD to 100G/lane 20km, 200G/lane 10km, and 400G/lane 3km," in *Optical Fiber Communication Conference (OFC) 2026* (Optica Publishing Group, 2026), p. Th1C.2.
- [19] S. Ran, Y. Guo, Y. Liu, T. Miao, Y. Wu, T. Qin, Y. Guo, L. Liu, Y. Zhu, Y. Li, Q. Zhuge, J. Chen, and L. Zhou, "A 4×256 Gbps silicon transmitter with on-chip adaptive dispersion compensation," *Nature Communications*, vol. 16, no. 1, p. 6268, 2025. DOI: <https://doi.org/10.1038/s41467-025-61408-7>.
- [20] H. Liu, K. Sozos, I. Teofilovic, S. Wantee, K. R.H. Bottrill, S. Malhouitre, S. Garcia, G. Sarantoglou, P. Bienstman, B. Charbonnier, C. Mesaritakis, C. Vigliar, F. Da Ros, A. Bogris, and P. Petropoulos, "200-Gbaud Single-Wavelength Direct-Detection Transmission over 75 km C-band SSMF Using a PIC-based Recurrent Spectrum Slicer," in *Optical Fiber Communication Conference (OFC) 2026* (Optica Publishing Group, 2026), p. M3B.6.
- [21] P. Zhu, Y. Yoshida, A. Kanno, and K. I. Kitayama, "Analysis and demonstration of low-complexity joint optical-electrical feedforward equalization (OE-FFE) for dispersion-limited high-speed IM/DD transmission," *Journal of Lightwave Technology*, vol. 41, no. 2, pp. 477–488, 2022. DOI: [10.1109/JLT.2022.3217287](https://doi.org/10.1109/JLT.2022.3217287).